

JOSE VILLA-ROMERO

Engineering Student (M2) – Intelligent, Secure and Communicating Systems
Software Engineering & Embedded Systems – Internship from February 2027

@josemvilla12@gmail.com

+33 7 43 54 90 42

Lille, France

jose-manuel-villa-romero

jmvilla12

jmvilla12-portfolio.netlify.app

Driving Licence (Car)



PROFILE

Final-year engineering student (M2 / Master's level) at IMT Nord Europe, specializing in Intelligent, Secure and Communicating Systems. Hybrid profile combining hands-on software engineering – currently a Software Developer Intern at LeanPay (Lille, France) – with a solid foundation in mechatronics and embedded systems (Mechatronics Engineering degree, Universidad Tecnológica de Pereira, Colombia). Looking for an engineering internship of at least 6 months, available from **15 February 2027**, in France or elsewhere in Europe, with the goal of transitioning into a full-time role (CDI/CDD) afterwards. Dual software and mechatronics background, with experience in a French startup and in a distributed international team.

PROFESSIONAL EXPERIENCE

Software Developer – Internship

LeanPay

Apr. 2026 – Aug. 2026

Lille, France

- Developed new TypeScript features for a B2B SaaS platform for accounts-receivable management (collections), reducing clients' Days Sales Outstanding (DSO).
- Onboarded new clients: general-ledger imports, PDF invoice parsing, data ingestion via FTP and email, each with its own configuration.
- Practiced TDD with Jest and designed a custom error-handling architecture: every import failure surfaces its real cause to both developers and clients. Version control on GitLab.

Software Developer

Alta AI | BlueSkyAI

Dec. 2024 – Dec. 2025

USA (remote)

- Built the entire frontend for JRVs, a multi-provider AI assistant (Chat-GPT, Gemini, Claude): React, TypeScript, Framer Motion animations optimized for performance.
- Implemented Firebase Authentication with two independent methods (Google/email and X/Twitter); translated Figma designs into production UI, working daily with the design team.

Mobile Developer – Freelance

CIDT (UTP) for Cinnov S.A.S.

Apr. 2025 – Jul. 2025

Colombia

- Built two Flutter features for an existing, live production app (a location-based social reporting network): a gamification system and a real-time map of nearby events.
- Consumed the backend team's NestJS API, tested with Postman; work tracked in Azure DevOps, with weekly check-ins with the client.

Engineering Assistant

MET GROUP SAS

Nov. 2021 – May 2022

Colombia

LANGUAGES

Spanish – Native

French – Bilingual (C1)

English – Bilingual (C1)

SKILLS

Web & Software Development

TypeScript JavaScript React
Vue Flutter Vite Tailwind CSS
Framer Motion Figma UI/UX

Backend & Cloud

Java Spring Boot Microservices
Node.js NestJS Python Django
Firebase SQL Docker Maven
REST API

Systems & Embedded

C C++ Qt6/QML CMake
MicroPython CAN Bus
MATLAB/Simulink Altium Designer

Cybersecurity & Networking

Stormshield Firewall VPN / SSL
NAT Network Filtering Linux
Bash

Data

Python pandas scikit-learn SQL
Power BI R

Tools & Methods

Git GitHub GitLab
Azure DevOps Scrum/Agile
TDD (Jest) Postman

Soft Skills

Autonomy Teamwork
Communication Self-Learning
Rigor Pragmatism

CERTIFICATIONS

- Built a CAN bus interface in MicroPython to retrieve vehicle telemetry (fuel level, speed, door status) and stream it to the cloud for real-time monitoring.
- Designed electronic prototypes and PCBs in Altium Designer.

Member, MECABOT Aeronautics Research Group

Universidad Tecnológica de Pereira

📅 Nov. 2019 – Jul. 2023 📍 Colombia

- Competed in engineering competitions (Aerodesign, Human-Powered Vehicle) and built lab automation solutions (Industry 4.0, robotics, aeronautics).

PROJECTS

E-Commerce Microservices Architecture

Student Project, IMT Nord Europe

📅 2025 – 2026 📍 Lille, France

- Multi-module Spring Boot backend (Catalogue, Cart and Tracking services, each with its own H2 database and REST API), API Gateway with Spring Cloud Gateway, Spring Data JPA/Hibernate, Maven build, React frontend (Vite + TypeScript + Tailwind CSS). Team of 3.

Agricultural VTOL Drone — Engineering Thesis

Universidad Tecnológica de Pereira

📅 2022 – 2023 📍 Colombia

- Designed (Autodesk Inventor, NACA 0012 airfoil) and additively manufactured (FDM 3D printing, PLA) a vertical take-off and landing drone; integrated an open-source Pixhawk 2.4.8 autopilot.

EDUCATION

Engineering Degree, Intelligent, Secure & Communicating Systems (SISC) — Master's (M2)

IMT Nord Europe

📅 Sept. 2025 – Present 📍 Lille, France

- M2 specialization: artificial intelligence, cybersecurity, cloud computing, embedded systems, advanced network architectures.
- Chosen managerial track: *"Developing the Business"* — commercial dimension of projects, goods and services.

Mechatronics Engineering — Bac+3 / Licence equivalent

Universidad Tecnológica de Pereira

📅 Jan. 2018 – Jul. 2023 📍 Colombia

- Two-stage degree (laddered "ciclos propedéuticos" structure) with two separately defended degree projects. Thesis: design and implementation of a VTOL drone for the agricultural sector, additive-manufactured airframe. GPA: 4.3/5.0.

- **Stormshield Network Administrator (CSNA)** — score 86%, valid until April 2029.
- **Junior Full Stack Developer** — Universidad Tecnológica de Pereira, 176h (2024).
- **Data Analysis** — Talento Tech bootcamp, Colombia (2024).
- **Python for Everybody** — University of Michigan, Coursera, 76h (2021).

AWARDS & HONORS

- **2026 — Winner, Challenge Entrepreneure, "Environment" category:** 1st place among about 40 teams (IMT Nord Europe, Polytech Lille, and business schools). Built a React/TypeScript app mapping the city's recycling collection points, in a team of 5, in one week.
- **2024 — Talent Seedbed Program (Pépinière de Talents), COLFUTURO:** selective program for high-potential academic students.
- **2023 — Distinguished Student in Mechatronics Engineering:** graduation GPA of 4.3/5.0 with no failed courses across the whole program.
- **2018–2023 — Cooeducar Scholarship** (academic merit).

VOLUNTEERING

- **Roman Guard (Garde Romaine)** (2016–2024): social and cultural project in Pereira, Colombia.
- **AeroKiddo** (2021, 2024): taught aeronautics fundamentals to children and teenagers (STEAM methodology), MECABOT group.

HOBBIES

Strength training, handball, soccer, cooking, music, partner dancing, language learning.